Fluids, Pressure and Buoyancy

Synopsis: Everything so far has been about solid objects – a block, a cart, a marble – things that keep their shape and that you can point at. A fluid does not keep its shape. Pour water into a different glass and it becomes a different shape; it is still the same water. So the questions we have been asking need rephrasing. "What force acts on it?" becomes awkward when the thing you are pushing spreads out under your hand.

The rephrasing is this: instead of force, ask about force per unit area – pressure. Instead of mass, ask about mass per unit volume – density. Both replace a quantity that depends on how much of the stuff you happened to scoop up with one that does not. A litre of water and a swimming pool of water have very different masses and the same density, and that is exactly why density is the useful number.

With those two, the whole of fluid statics follows from one idea: the fluid above you has weight, and it is pressing down on you. Go deeper and there is more of it above you, so the pressure rises. That single sentence gives you Pascal's principle, the hydraulic jack that lifts a car with one hand, and Archimedes' principle – why a steel ship floats while the same steel, melted into a lump, goes straight to the bottom. Note 2 then sets the fluid moving.

Performance Expectations: Define density and pressure and use them. Explain where the pressure in a fluid comes from, and find the pressure at a depth. Use Pascal's principle to solve a hydraulic problem. Use Archimedes' principle to decide whether something floats, and to find a density by weighing an object twice.

For PHY 151 — with calculus: Get the depth result as the solution of a differential equation rather than as a formula, and see what has to change when the density is not constant.

Before you read on – predict. A steel ship floats. Melt that same ship down into a solid lump of steel and drop it in – it sinks. Same steel, same mass, same water. Commit to one sentence here saying what changed, before you watch. You will come back to it at the end.

What changed, in one sentence

Instructions and Questions

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1) What a fluid is, and why density replaces mass.
   A fluid is anything that flows – anything whose molecules are free to move past one another. That takes in liquids and gases, which is worth saying out loud, because air behaving like a fluid is what makes half the examples in Note 2 work. Write your own definition of density first; it stays yours. Then compare.

Your definition of density	**Physics definition (with units)**

The one conversion to have at your fingertips – box it where you can find it again:

ρ_{water} = _____ g/cm^3 = _____ kg/m^3

1 kg/m^3 = _____ g/cm^3

i) Water is the reference for everything in this note. Above what density does a substance sink in water, and below what density does it float?
ii) Why is that a statement about density and not about mass? One sentence.

2) Pressure – force spread over an area.
   Push on a fluid and the push spreads out. What matters to the fluid is not how hard you pushed but how hard you pushed per unit of area. Write the equation, its units, and the name of the unit:

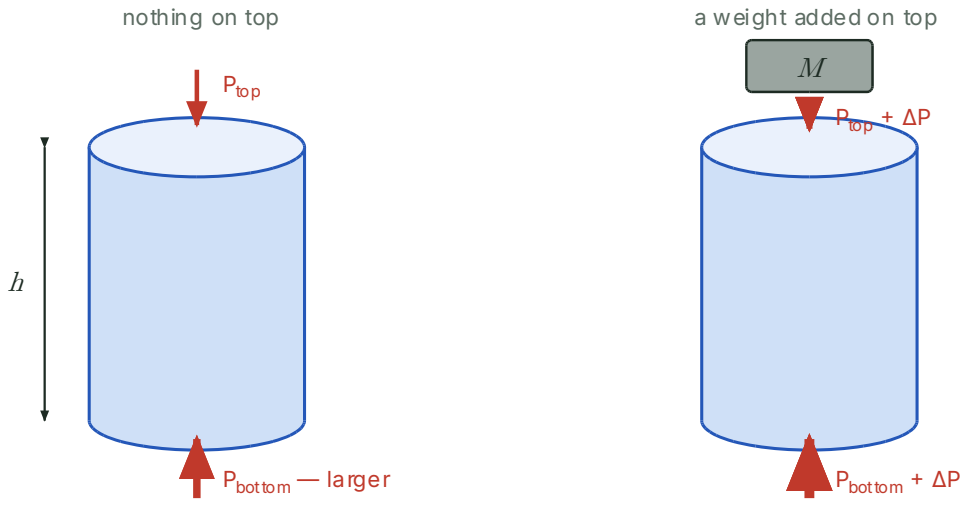
P = _____ F in _____ , A in _____ , so P is in _____ , which has its own name: the _____ .

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Read the fraction. There are two ways to get a big pressure, and they are different ways. One line each on whether it is F or A doing the work:

Situation	**Why the pressure comes out the way it does**
A pin prick hurts	
A poke with your fingertip does not	
An elephant stands on soft ground without sinking	

3) Where the pressure in a fluid comes from.
Nothing is pushing on a glass of water sitting on the table, and yet the pressure at the bottom of it is higher than the pressure at the top. The reason is the only one available: the water above has weight, and it is resting on the water below.



Two separate facts, and keeping them apart is most of this section. Mark them on the figure: pressure depends on height – the gap between the two red arrows on the left. A change in pressure does not – press on the top and every point rises by the same ΔP, as on the right.

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Build the depth formula from the weight above. Take a column of fluid of height h sitting on an area A, and fill in each line as the video works through it:

P = weight above / area = _____

and m = _____ , so P = _____ = _____

but V/A is just the _____ of the column [$m^3 \div m^2 = m$]

P = _____ the pressure the fluid itself adds, at depth h

i) Notice what is not in that result. Which quantity cancelled, and what does its absence tell you about a narrow pipe compared with a wide lake at the same depth?

One more thing is sitting on top of the water: the atmosphere. So the absolute pressure at a depth is what the fluid adds plus what the air was already contributing.

P_{abs} = _____ with P_{atm} = _____ Pa at sea level

How deep do you have to go in water before the pressure is two atmospheres?	**Your working**

i) Say in one sentence what ten metres of water is worth, in atmospheres. Then say what twenty metres is worth.
ii) A mercury barometer is 760 mm tall. Mercury is 13.6 times denser than water. How tall would a water barometer have to be?

The same result, as a differential equation.

Take a slab of fluid of thickness dy at height y. The pressure difference across it supports the slab's own weight, which gives $dP/dy = -\rho g$ – minus, because pressure falls as you go up. Integrate with ρ

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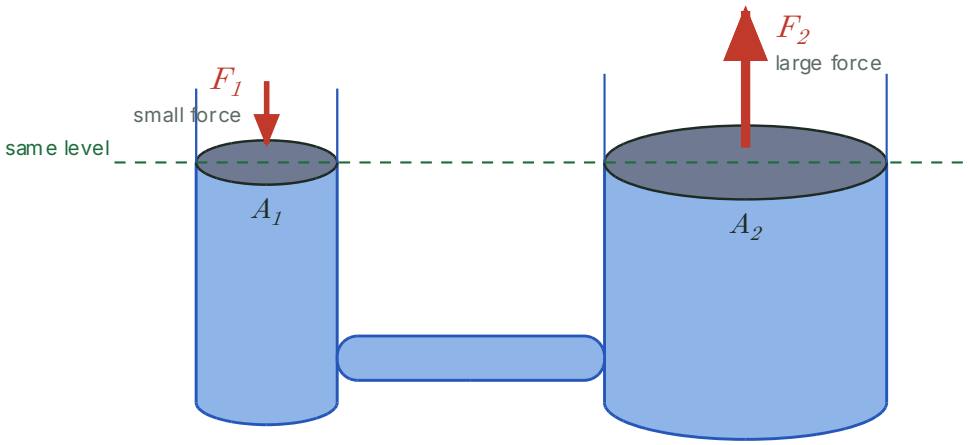
constant and you recover $P = P_0 - \rho g y$, which is the formula above with the sign absorbed into "depth" instead of "height". Written this way it is obvious what the constant-density assumption was doing: for a gas, ρ is itself a function of y, the integral no longer collapses to a product, and you get an atmosphere whose pressure falls off roughly exponentially rather than linearly.

i) Integrate $dP/dy = -\rho g$ from the surface down to depth d and confirm you get $P_0 + \rho g d$.

ii) If instead $\rho = \rho_0 e^{-y/H}$, carry out the integral and say what P(y) looks like.

4) Pascal's principle, and the hydraulic jack.

A change in pressure applied to an enclosed fluid is transmitted undiminished to every part of it. Undiminished is the word that matters – the increase does not fade with distance. The pressure itself still depends on height, but the change does not. That is a statement about fluids. It is also a machine.



One fluid, one pressure. Four times the area, four times the force.

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Mark A_1 and A_2 on the sketch, then write the two lines – and write the reason next to the first one, because the reason is the whole trick:

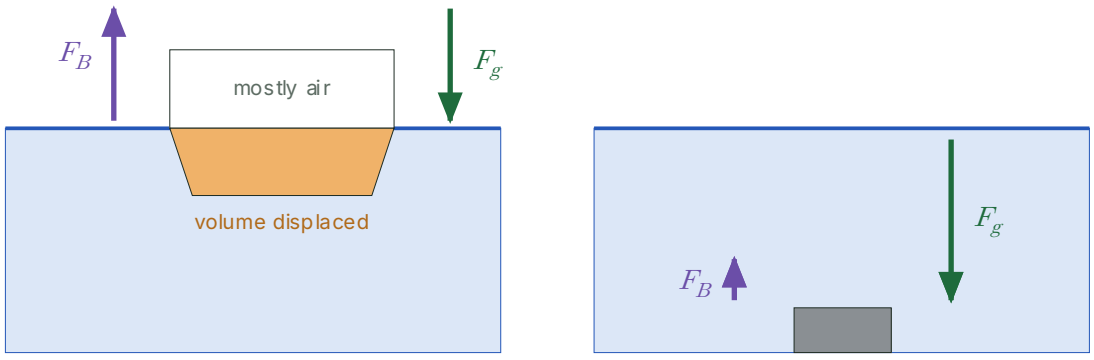
$P_1 = P_2$ because _____
_____ = _____ → F_1 = _____

i) In your own words: what does a hydraulic jack multiply, and what does it not multiply?
ii) A jack has $A_2 = 4A_1$. You push down with 150 N. What force comes up on the wide piston?
iii) To raise the wide piston by 1 cm, does the narrow piston move more than 1 cm, less, or the same? Why? (Think about where the fluid goes.)

5) Archimedes' principle – why the ship floats.
Put something into a fluid and it has to shove some of that fluid out of the way. The fluid pushes back, and Archimedes' principle says exactly how hard. Write it in words first, then in symbols, and watch the subscripts – neither one of them is a property of the object:

F_B = _____ (in words)
$F_B = W_{w,dis} = M_w g$ = _____

a hull full of air — the same metal, melted down



floats — the two balance — sinks — weight wins

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Mark the forces on both panels. Same steel, same mass, same water – and the arrows are drawn to the same scale in both. Then derive the floating condition: a floating object is in equilibrium, so the buoyant force exactly carries its weight.

$F_{g,ship} = F_B \quad \rightarrow \quad M_{ship}g = $ _____

$\rho_s V_s g =$ _____ $\rightarrow \quad \rho_s V_s =$ _____

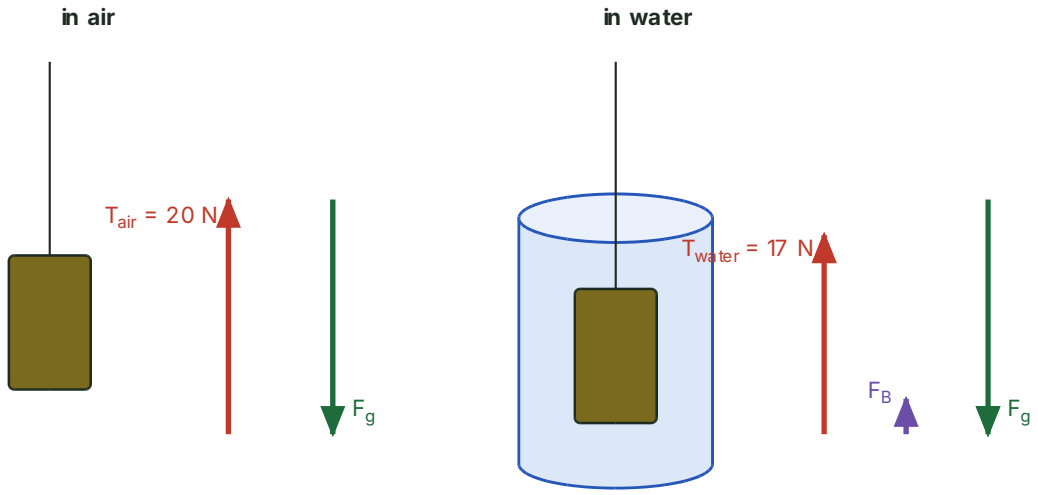
i) A hull full of air floats and a solid lump of the same steel does not. Which quantity in the line above is different for the two, and which is the same?

ii) For a **solid** object, $V_{dis} = V_s$. Put that into the condition and say what it collapses to. Then compare it with the rule you wrote in question 1.

iii) Now answer your prediction from the top of this handout. Was the sentence you committed to right? If you said "the shape changed", say why shape matters.

6) Weighing something twice to find its density.

Here is the part of Archimedes that earns its keep in the laboratory, and it is what you will do in the Buoyancy Lab. Hang a lump of metal from a spring scale and read the tension. Then lower it into water, fully submerged, and read the tension again. It weighs less – not because its weight changed, but because the water is now helping to hold it up.



The weight is the same in both. The scale reads less because the water carries part of it.

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Draw the two free-body diagrams side by side. In air there are two forces; submerged there are three. Note that the two weight arrows are the same length, because the weight has not changed – what the scale reports is the **apparent** weight.

From the two scale readings, find the density of the water	**Your working**
T_{air} = 20 N, T_{water} = 17 N	

Then do it in symbols, and watch what cancels:

$T_{water} + F_B = T_{air}$ → F_B = _____ = _____ N

fully submerged, so the water displaced has the metal's own volume V, the same V on both sides

$\rho_w Vg = F_B = T_{air}(1 - T_w/T_a)$ and $T_{air} = M_m g$ = _____

ρ_w = _____ what cancelled? _____

i) With the numbers above, work out ρ_m if the water is the known 1000 kg/m^3.
ii) Run the method the other way – known water, unknown sample. What have you measured, and what did you never have to measure?
iii) Why does that matter for an irregular lump?

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Check yourself

Four to do once you have filled the notes in. Work them here, then check them against the interactive version at the bottom of the note page.

1) A block measuring 2.0 cm × 3.0 cm × 5.0 cm has a mass of 81 g. What is its density, in kg/m^3?

2) You press down on the piston of a sealed cylinder of water. The pressure increase at a point near the bottom, compared with the increase near the top, is
   i) larger, because pressure grows with depth
   ii) the same – a change in pressure is transmitted undiminished
   iii) smaller, because the water above absorbs some of it
   iv) impossible to say without the cylinder's area

Why?

3) The buoyant force on a fully submerged object depends on
   i) the mass of the object
   ii) the density of the object
   iii) the density of the fluid and the volume displaced
   iv) the depth it is held at

Why?

4) A sample hangs from a scale reading 12.0 N in air and 9.0 N fully submerged in water. The buoyant force on it is
   i) 21.0 N
   ii) 3.0 N
   iii) 9.0 N
   iv) 12.0 N

Why?

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Your notes

Space for the sketches, the derivations you want to redo yourself, and anything from the video that did not fit above. Everything on this handout is fluid standing still; Note 2 sets it moving. The method of the last section is what you carry into the Buoyancy Lab.

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